

Qingsen Zhang

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RESEARCH EXPERIENCE

Ph.D. Candidate, Computer Science Binghamton University

Aug. 2024 – Present

Binghamton, NY

Advisor: Prof. Yincheng Jin

Wearable 3D Face Reconstruction Doctoral Research Project

- **Data collection:** Designed and operated a dual-rim wearable capture system; collected and processed more than 700K synchronized frame pairs from 17 Deaf ASL signers and generated template-aligned 3D supervision from frontal reference videos.
- **Pipeline design:** Developed an end-to-end rim-to-canonical reconstruction pipeline that maps fragmented near-field dual views to complete, template-aligned 3D face meshes using multi-scale view conditioning and rectified flow in UV displacement space.
- **Training and evaluation:** Implemented and trained the models in PyTorch, reproduced and adapted comparison methods, and evaluated surface and facial landmark reconstruction on participant-disjoint data splits.

Egocentric Stereo ASL Hand Dataset Dataset Curation and Annotation Quality Assurance

- Contributed to collecting and organizing head-mounted stereo recordings from 25 Deaf native ASL signers; took a primary role in processing, reviewing, and quality-checking linguistic and 3D hand-pose annotations for continuous signing data.

INDUSTRY EXPERIENCE

Machine Vision Engineer Wuxi Lead Intelligent Equipment Co., Ltd.

2021

Wuxi, China

- **Vision system development:** Developed C++ machine-vision software with OpenCV for engineer-to-order lithium-ion battery manufacturing equipment, including dimensional measurement, surface-defect inspection, electrode-tab positioning, and geometric calibration. Adapted the software across machine configurations and maintained HALCON-based systems where applicable.
- **On-site commissioning:** Spent approximately three months at customer manufacturing sites commissioning equipment, calibrating and tuning vision systems, diagnosing production issues, and modifying software as requirements emerged during staged delivery.

Lecturer, School of Blockchain Jiangxi University of Software Professional Technology

2022

Nanchang, China

- Designed and taught introductory Python and blockchain courses to approximately 300 undergraduate students across six classes.
- Organized competition preparation and advised student teams on foundational computing skills and applied data-analysis challenges.

EDUCATION

Ph.D. Candidate in Computer Science Binghamton University

Aug. 2024 – May 2030 (expected)

Binghamton, NY

Master of Engineering in Computer Science Virginia Tech

Aug. 2023 – May 2024

Falls Church, VA

M.S. in Computer Science University of Delaware

Sep. 2019 – Dec. 2020

Newark, DE

B.S. in Computer Science University of Delaware

Sep. 2016 – Aug. 2019

Newark, DE

PUBLICATIONS

- Y. Cai, Z. Li, T. Lu, Y. Zhu, Y.-S. Wu, **Q. Zhang**, X. Xu, Z. Jin, M. Gowda, and Y. Jin. "Toward Scalable ASL Education: Egocentric Stereo Sensing with LLM Feedback for Error-Aware Learning." *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2026.
- **Q. Zhang**. "Attendance System Based on Blockchain and Face Recognition." *2022 International Conference on Smart Applications, Communications and Networking (SmartNets)*, 2022.

TECHNICAL SKILLS

Programming and Tools: Python, C++, PyTorch, PyTorch3D, CUDA, OpenCV, HALCON, Linux, Git, Docker